Software Design Document

Learning Analytics Prediction System (Ch07)

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Abstract

This Software Design Document (SDD) details the architecture and implementation of the "Learning Analytics Prediction System" (Ch07). The system is a machine learning pipeline designed to predict student academic performance in online courses based on their interaction logs from the Moodle Learning Management System (LMS). By analyzing behavioral patterns such as study session regularity (entropy) and engagement intensity, the system identifies at-risk students week-by-week, enabling timely instructor interventions.

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1. Application Architecture

The system follows a modular pipeline architecture designed for batch processing of student data.

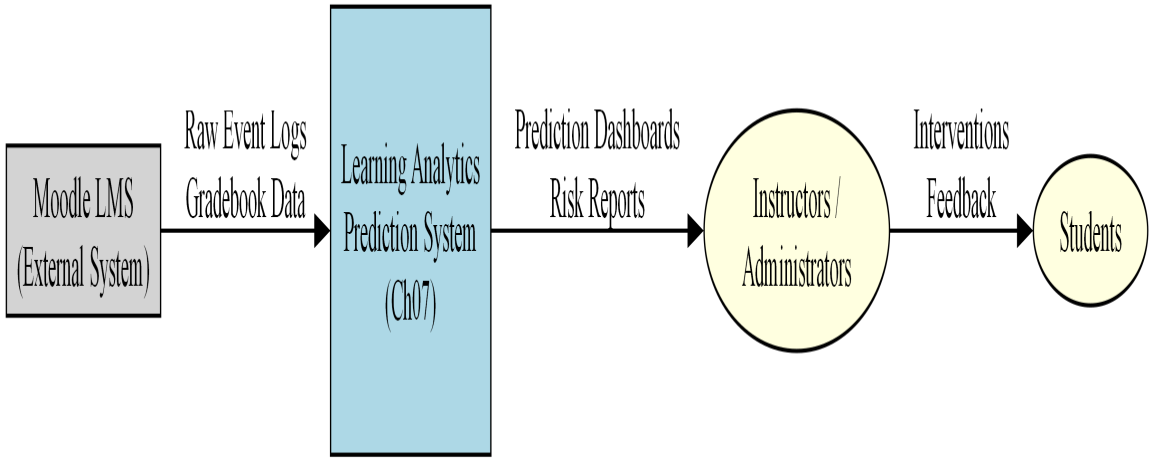

Figure 1: System Context Diagram

1.1 Core Components

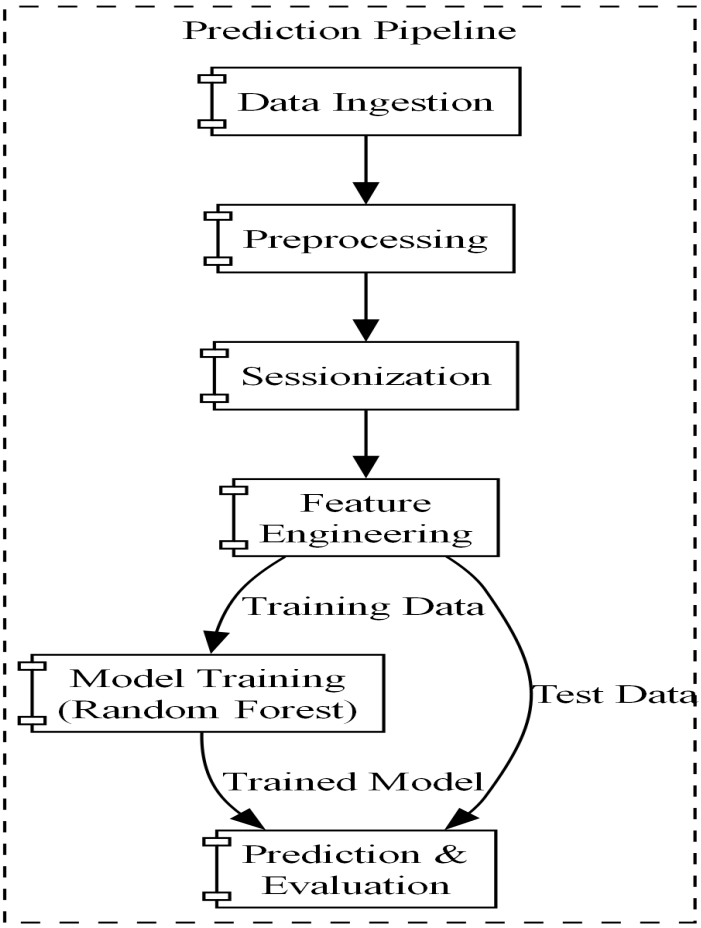

Figure 2: Component Diagram
The system consists of six primary components:

- **1. Data Ingestion:** Parsers for reading raw `Events.xlsx` and target `Results.xlsx`.
- **2. Preprocessing:** Timezone normalization and data cleaning.
- **3. Sessionization:** Algorithms to group raw clicks into study sessions (1.5h threshold).
- **4. Feature Engineering:** Computation of 20+ predictive features per student/week.
- **5. Model Training:** Random Forest classifiers/regressors trained on historical weeks.
- **6. Prediction Engine:** Applies models to generate risk scores for current students.

2. Data Model Schema

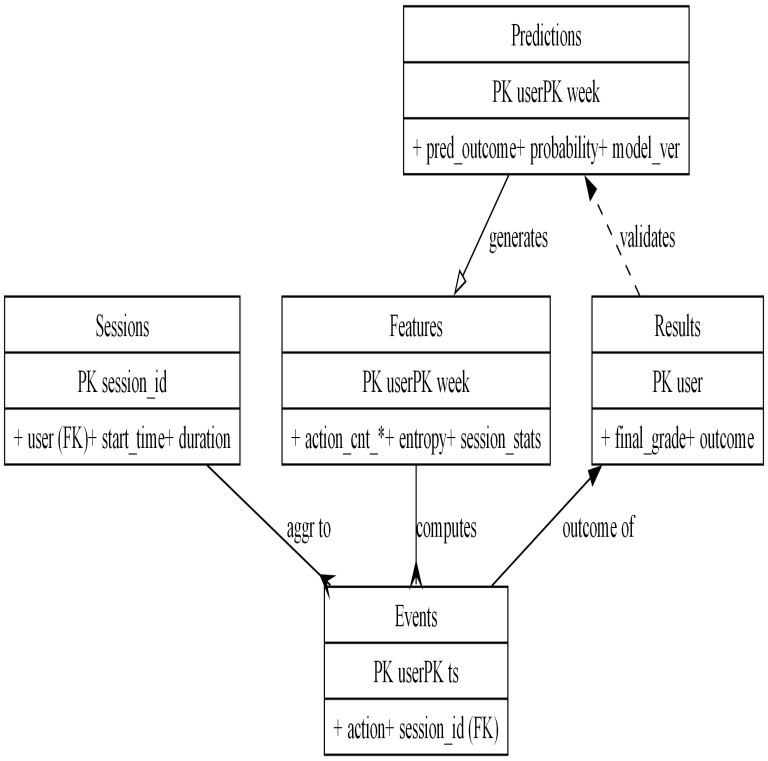

Figure 3: Entity-Relationship Diagram

2.1 Entity Definitions

Events Table (Fact)

Column	Type	Description
user	string	Unique Student ID
ts	datetime	Interaction Timestamp
action	string	Interaction Type (view, post)
week	int	Course Week (1-5)

3. User Interface

3.1 Input/Output Interfaces

The system is primarily a backend service processing file-based inputs:

- **Inputs:** Excel files (`Events.xlsx`, `Results.xlsx`) provided by course admins.
- **Outputs:** CSV files (`predictions.csv`) for dashboard consumption.

3.2 Conceptual Dashboard

The prediction results feed into an Instructor Dashboard featuring:

- **Risk Heatmap:** Visualizing student risk levels across weeks.
- **Detail View:** Drill-down into a student's specific feature drivers (e.g., 'Low Session Consistency').

4. Design Decisions

Sessionization Threshold (1.5h): Chosen based on educational psychology research indicating standard study block limits.

Random Forest Model: Selected for its ability to handle non-linear relationships and high-order interactions in behavioral data better than linear models.

Entropy Features: Used explicitly to capture the 'consistency' of student habits, distinguishing between crammers (low entropy) and regular studiers (high entropy).

5. Algorithm & Methodology

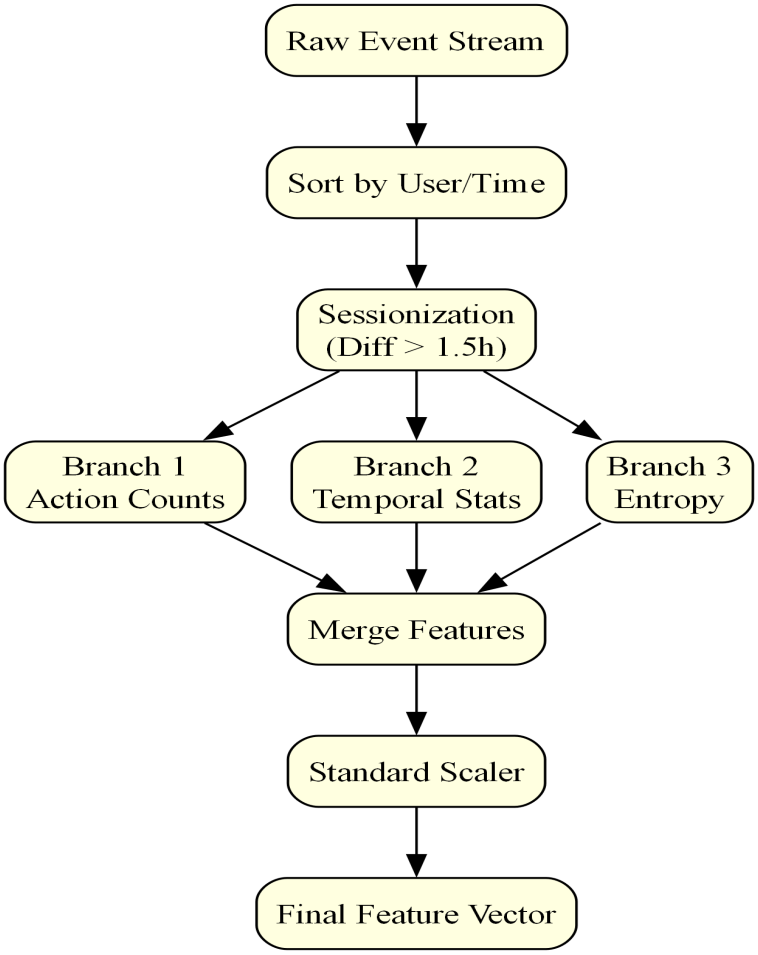

Figure 4: Feature Engineering Pipeline

5.1 Feature Engineering

Key features include:

- **Activity Counts:** Total actions, actions per type.
- **Temporal:** Median session length, average gap between active days.
- **Entropy:** Shannon entropy of daily activity counts.

5.2 Data Flow Logic

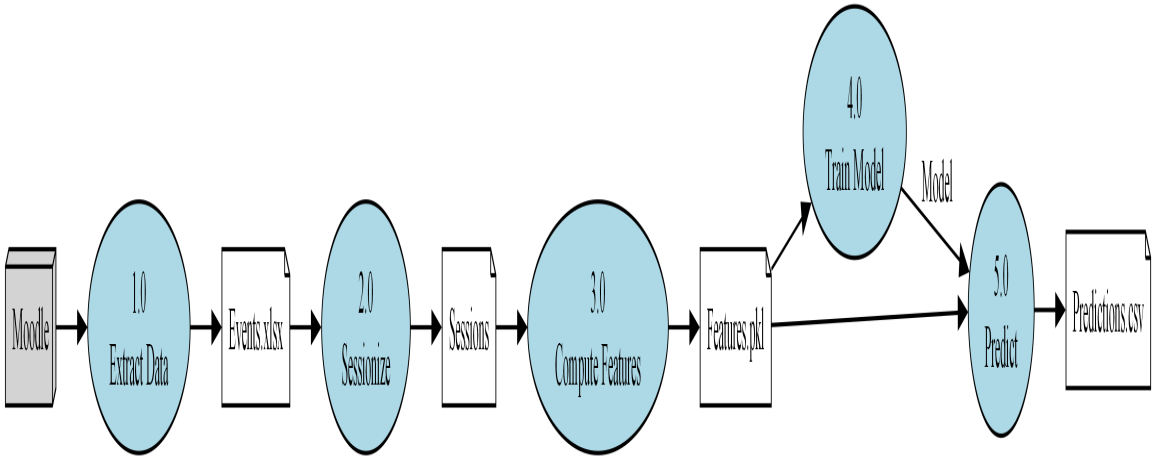

Figure 5: Data Flow Diagram (Level 1)

6. Security & Data Privacy

The system is FERPA-compliant. All internal processing uses pseudonymized User IDs. Mapping tables are stored offline. Data access is restricted to authorized instructors only.

7. Deployment & Operations

The system operates on a weekly batch cycle (Sunday nights). Models are versioned (e.g., v1.0, v1.1) to track performance changes over the semester.

8. Testing Strategy

Figure 6: Expected Model Performance (Simulated)
Validation ensures F1-score > 0.65 before model deployment.

Appendix A: Feature Dictionary

Feature	Description
action_cnt	Total interactions
session_cnt	Total study sessions
entropy_daily_cnts	Regularity of daily login patterns
median_session_len	Typical study duration

Appendix B: Configuration

```
RF_N_ESTIMATORS = 200
```

Appendix C: Glossary

- **Sessionization:** The process of grouping raw events into meaningful study periods.
- **Entropy:** A statistical measure of randomness used here to proxy study consistency.
- **FERPA:** Family Educational Rights and Privacy Act.